

Instructions: 1. Attempt any Five Questions

2. Each questions carries equal marks

- Q.1 a) An analog signal $x_a(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per second
- (i) Determine the Nyquist sampling rate for $x_a(t)$
- (ii) What are the frequencies in radians, in the resulting discrete time signal $x[n]$?
- b) Define Continuous Wavelet Transform. State and prove any three properties of the same.
- Q.2 Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x[n]$. Perform the following calculations without explicitly evaluating $X(e^{j\omega})$:
- a) Evaluate $X(e^{j0})$ b) Find phase of $X(e^{j\omega})$
- c) Evaluate $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$ d) Find $X(e^{j\pi})$
- e) Evaluate $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$
- For given sequence, $x[n] = \{1, 2, 4, 2, 4, 2, 1\}$
- Q.3 a) Consider the following LTI System with impulse response $h(n) = \alpha^n u[n]$ with $|\alpha| < 1$ and suppose that the input to this system is $x[n] = \beta^n u[n]$ with $|\beta| < 1$. Find the output $y(n)$ of this system. (06)
- b) Draw the direct form-I and direct form-II structure for causal LTI system described by following difference equations
- i) $y[n] = 4x[n] + 3x[n-1] - 2x[n-2]$
- ii) $y[n] = \frac{1}{3}y[n-1] - 5x[n-1]$
- Q.4 a) The following is the Fourier transform of discrete-time signal. Determine the signal corresponding to that transform.